

Case: Educational Performance

Layout: The case should follow these layout guidelines:

- Maximum 3 pages excluding figures and tables.
- Font size 12; 1.5 spacing, standard margins; numbers rounded to at most 3 decimals.
- Two regression tables: one for Part 1 and one for Part 2.

Points: This case has a maximum of 100 points allocated as follows:

- Part 1: 30 points.
- Part 2: 30 points.
- Figures: proper and correct formatting, 10 points.
- Tables: proper and correct formatting, 10 points.
- Clear and concise writing, quality, and overall layout: 10 points.
- Clean R script, included as an appendix: 10 points.

Grading: Each component is graded on a 1-6-10 scale:

- 1: nothing or very incomplete solution.
- 6: complete submission but significant inaccuracies or imprecisions.
- 10: serious effort and mostly correct.

Use of AI: AI may be used as a coding helpdesk and to improve writing, for example as a spell-check or readability tool. It may not be used to generate answers. Students remain wholly responsible for submissions, which should represent independent work.

If overreliance on AI is suspected, the group may be asked to prepare for an oral exam covering the case, code, and results. The grade may then be based on the oral exam. Evidence of using AI to do the work, such as prompts asking AI to interpret results, may result in a fail.

Introduction

The Programme for International Student Assessment (PISA) 2022 is a large-scale international survey that evaluates the knowledge and skills of 15-year-old students worldwide. It focuses on domains such as mathematics, reading, and science, while also collecting background information on students, schools, and learning environments. The data helps researchers and policymakers understand educational outcomes and inequalities across countries. In this assignment, students use the PISA 2022 dataset to explore empirical questions using real-world educational data.

The dataset and variable overview are available through the course learning environment. Each group uses data from a different set of 8 countries, as indicated in the accompanying variable/country allocation file.

PART 1

In Part 1, investigate how educational success in one area relates to educational success in another subject.

Question 0: Data Preparation

First, select the correct set of countries. Standardize the three test scores: MATHSCORE, READINGSORE, SCIENCESORE. If X is a random variable with sample mean $\bar{X}$ and standard deviation σ_X , the standardized variable is:

$$X_{\text{standard}} = (X - \bar{X}) / \sigma_X$$

Report the summary statistics, mean and standard deviation, of the original and standardized test scores. What units are the standardized variables? What is the advantage of standardization?

Question 1: Univariate Data Visualization

Draw one graph illustrating the distribution of the three standardized test scores. Describe and interpret.

Question 2: Bivariate Data Visualization

Draw a graph illustrating the correlation between standardized math and standardized reading scores. Discuss and interpret.

Question 3: Univariate Regression

Formally test the relationship established in Question 2 through a regression analysis. Use the standardized reading score as the independent variable. Adjust standard errors to allow for heterogeneity in this regression and in all following regressions. Discuss the two estimates, including interpretation and significance. Mention the test, hypothesis, and decision rule for significance once. Do you think this relationship is causal? Justify your answer.

Question 4: Multivariate Regression

Next, consider whether reading score depends on the language the student speaks at home. Repeat the regression from Question 3, now also including whether the student mostly speaks the test language at home, measured by ST022Q01TA. Transform the language variable before using it. If there are missing values, exclude those observations. Interpret the results, including significance.

Question 5: Interaction

Now hypothesize that the student's main language may also affect the correlation between math and reading scores. Test this prediction in a regression. Also illustrate the heterogeneous correlations in a graph. Describe and discuss the result.

Table: Part 1

Produce one table with all regression results from Part 1. Format it properly, as in a journal article. The table should be understandable to someone unfamiliar with the dataset.

PART 2

In Part 2, analyze the effect of circumstances and facilities at the student's home, such as ICT resources, on test scores.

Question 6

Start by regressing the standardized math score on ICTRES, a measure of ICT resources at home. Also include dummies for each country in the dataset:

$$\text{MATHSCORE}_i = \beta_0 + \beta_1 \text{ICTRES}_i + \sum_C \gamma_c I(\text{Country}_i = C) + u_i$$

where $I(\text{Country}_i = C)$ is a dummy variable equal to 1 if student i is from country C and 0 otherwise. Provide a precise interpretation of the coefficient for ICTRES.

Question 7

Estimate the same regression model again, but now use log-transformed math test scores, not the standardized version, as the dependent variable.

- Provide a precise interpretation of the coefficient for ICTRES.
- Provide a precise interpretation of the coefficient for one country dummy.
- Briefly comment on the benefits and downsides of log-transformed test scores compared with the standardized variable in the previous question.

Question 8

The effect of ICT resources at home on test scores is likely to be confounded by other omitted factors in the home environment. Control for whether the family lived in poverty, using ST258Q01JA as a proxy. This variable measures how often there was insufficient food in the last 90 days. Add this variable to the regression model as a control variable, considering that it is categorical. Use the standardized test score as the dependent variable, not the log-transformed variable. Answer:

- Interpret the effect of poverty/lack of food. Is it in line with expectations?
- Comment briefly on how the coefficient for ICTRES changes after adding the control variable.
- Calculate the omitted variable bias in the coefficient of ICTRES in the regression without the control variable.
- Use the omitted variable bias formula to show whether the correlation between ICTRES and lack of food is positive or negative, without computing it from the data.

Question 9

There is likely still omitted variable bias, even when including the proxy for poverty. As a final step, consider instrumental variable estimation using the highest level of education attained by the parents, HISCED, as an instrument for ICT resources at home. Before estimating the model, answer:

- State precisely the relevance condition in this case, and discuss briefly whether it is satisfied.
- State precisely the exogeneity condition in this case, and discuss briefly whether it is satisfied.

Question 10

Estimate the effect of ICT resources at home on test scores using HISCED as an instrument. HISCED is categorical, so include it as a set of dummies. The ivreg function from the AER package may be used.

- Provide a brief interpretation of the coefficient for ICTRES. Does it have a causal interpretation?
- If the instrument is assumed valid, conclude whether the omitted variable bias in the OLS model was positive or negative. Explain your answer.

Table: Part 2

Produce one table with all regression results from Part 2. Format it properly, as in a journal article. The table should be understandable to someone unfamiliar with the dataset.

Include the R script as an appendix.